WSABI Summer Research Lab **Summer 2026**

Lab 9: The Bootstrap

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9.1 Expected Points Bootstrap

In Lab 5, you built an expected points model for NFL play-by-play data using `05_expected-points.csv`. In this lab, you will return to that model and use the bootstrap to quantify uncertainty for a fitted expected points value.

Use your preferred expected-points model from Lab 5. If you want a concrete default, a reasonable choice is a multinomial model for `pts_next_score` with:

- a spline in `yardline_100`,
- `down` treated as categorical,
- `ydstogo` on a log scale or through a spline,
- and `half_seconds_remaining` entered flexibly.

Take the target estimand to be expected points for a `1st-and-10` at the opponent 35 with `ydstogo = 10` and `half_seconds_remaining = 1800`. If your model also includes `posteam_spread`, set it to 0.

9.1.1 Your Task

1. Refit your preferred expected-points model from Lab 5 and compute the fitted expected-points value at the target state above.
2. Decide which bootstrap variation is most appropriate here: observation bootstrap, cluster bootstrap, parametric bootstrap, or residual bootstrap. State your choice and explain why it is the best match for this dataset.
3. Implement that bootstrap method with at least $B = 200$ bootstrap resamples. For each resample, refit the model and recompute the target expected-points value.
4. Plot the bootstrap distribution of the target expected-points value.
5. Report a bootstrap standard error and a 95% percentile interval for the target expected-points value.
6. Briefly explain why a naive row-by-row observation bootstrap is less appropriate here.

9.2 Bootstrap for Free Throws

In Lab 8, you constructed Wald and Agresti-Coull confidence intervals for NBA free-throw percentages and studied their coverage. Now, use the bootstrap on that same problem.

9.2.1 Your Task

1. Use the bootstrap to construct a 95% confidence interval for the true free-throw percentage of NBA players.
2. Overlay the bootstrap interval on the player plot you made in Lab 8. Compare the bootstrap interval to the Wald and Agresti-Coull intervals.
3. Use the bootstrap to construct a 95% confidence interval for the simulation study you performed in Lab 8.
4. Add a plot of the coverage probability for the bootstrap confidence interval. How does it compare to the Wald and Agresti-Coull intervals?

We will return to the bootstrap again in the Machine Learning unit.